

LangFIR: Discovering Sparse Language-Specific Features from Monolingual Data for Language Steering

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Introduction

- Background:** Reliably controlling the language of an LLM's outputs remains difficult. Representation steering addresses this by adding a language vector to the residual stream at inference time.
- Problem:** Finding these directions usually requires *multilingual or parallel data*, which can be expensive to obtain.
- Question:** Can we find language-specific directions from monolingual data alone?

Key Insight

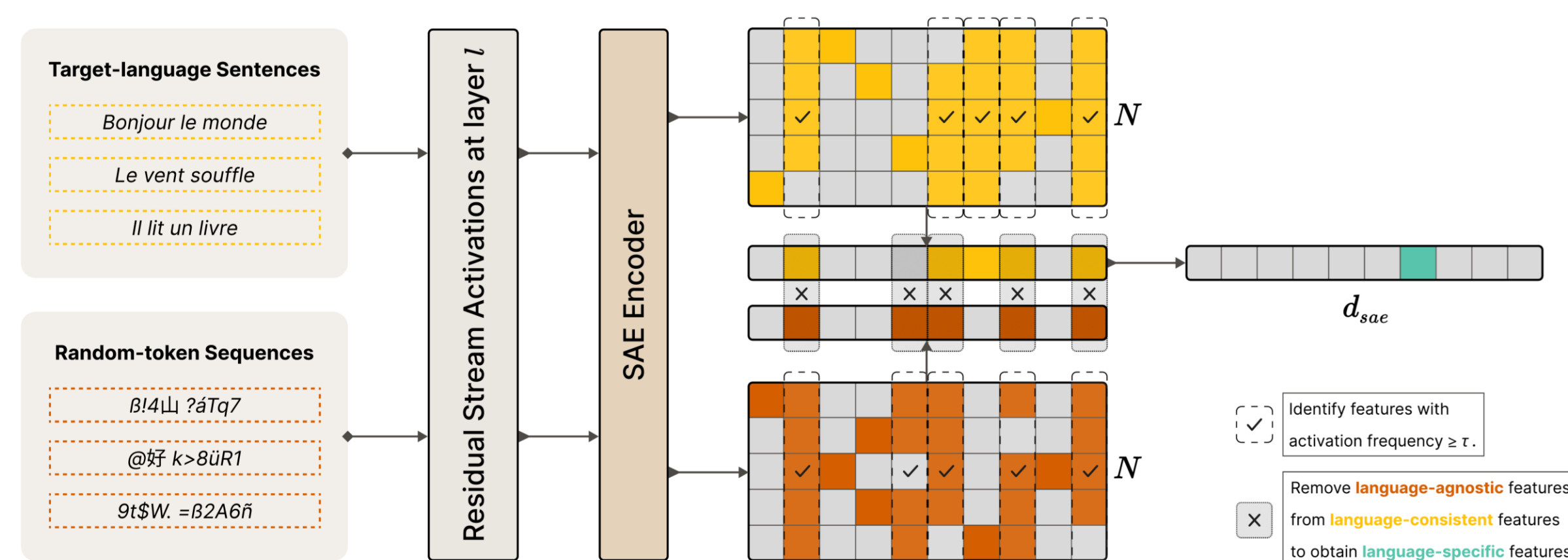
- Not every feature that fires on target-language text encodes language.
- Many respond to language-agnostic structures such as punctuation, whitespace, and symbol patterns.
- Hypothesis:** Random-token sequences (strings sampled uniformly from the tokenizer's vocabulary) lack coherent linguistic content, so they provide a simple, data-free proxy for surfacing these language-agnostic features.

A Feature That Isn't About Language

TOP ACTIVATIONS
4.22 { name : 'a malformed account ar
4.16 1998 The Hoy den
4.13 production are the documentaries 'THE STORY OF A
3.92 Amy gd ala Magazine Ivan Inf ante Returns
3.88 Progress Tag Archives Organ Donation Book

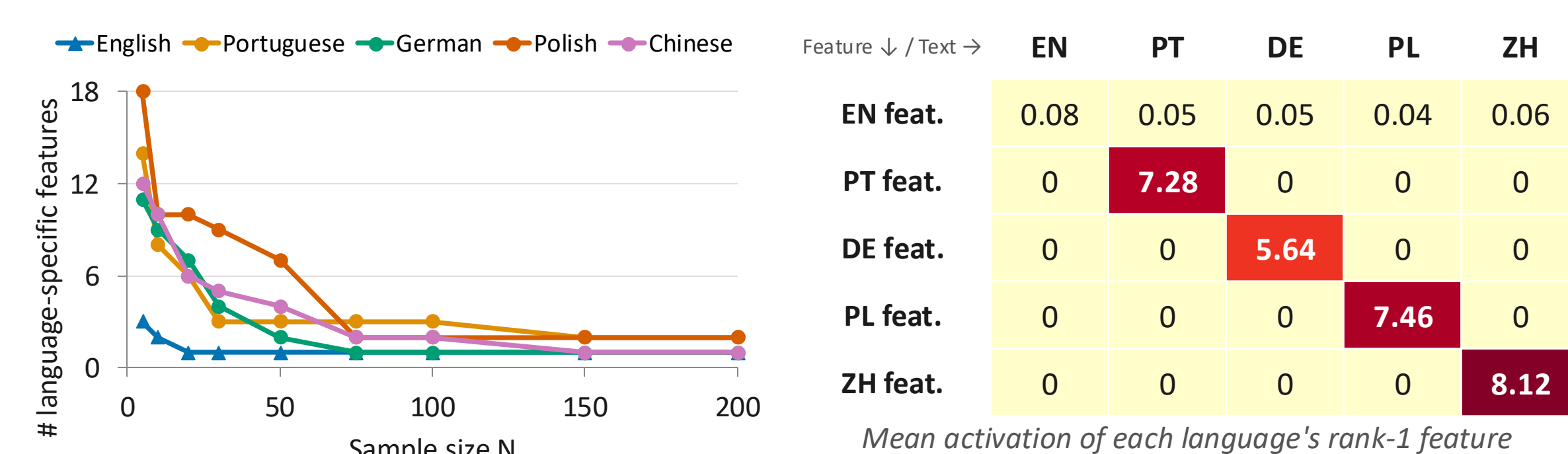
Top activations of SAE feature 3157 (Llama 3.1 8B, layer 18, Neuronpedia). It fires on dashes, quotes, and colons.

Method Overview

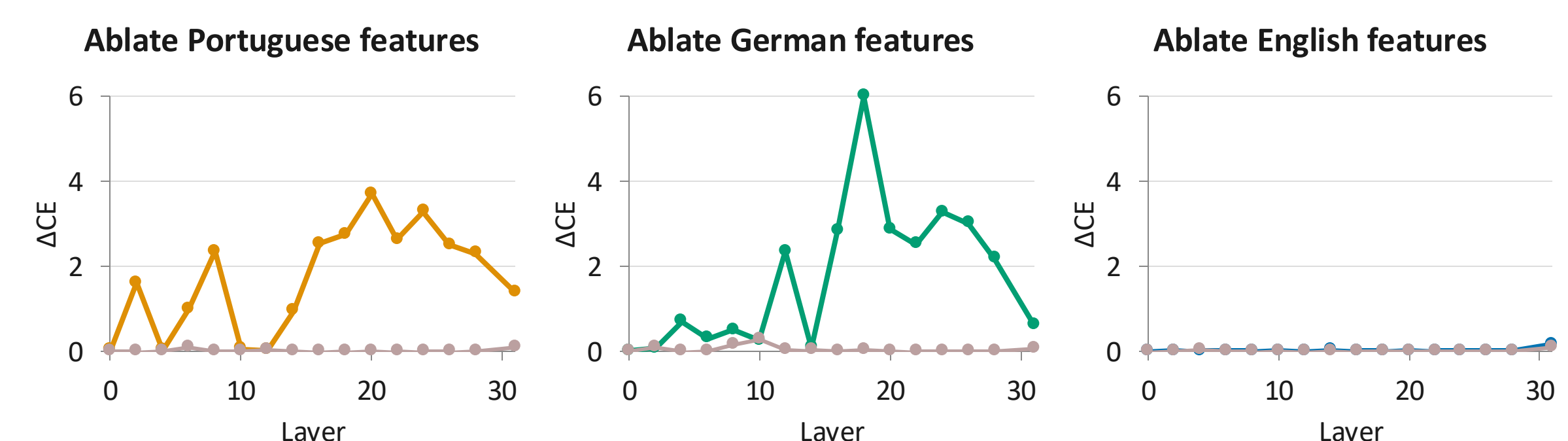


- Pair each target-language sentence with a random-token sequence of similar length.
- Extract residual activations at layer l and encode them through an SAE.
- For each input type, keep features that are active in at least a fraction τ of samples.
- Remove **language-agnostic** features from **language-consistent** ones to get a sparse set of **language-specific** features.

Language-Specific Features: Sparse, Selective, Causal



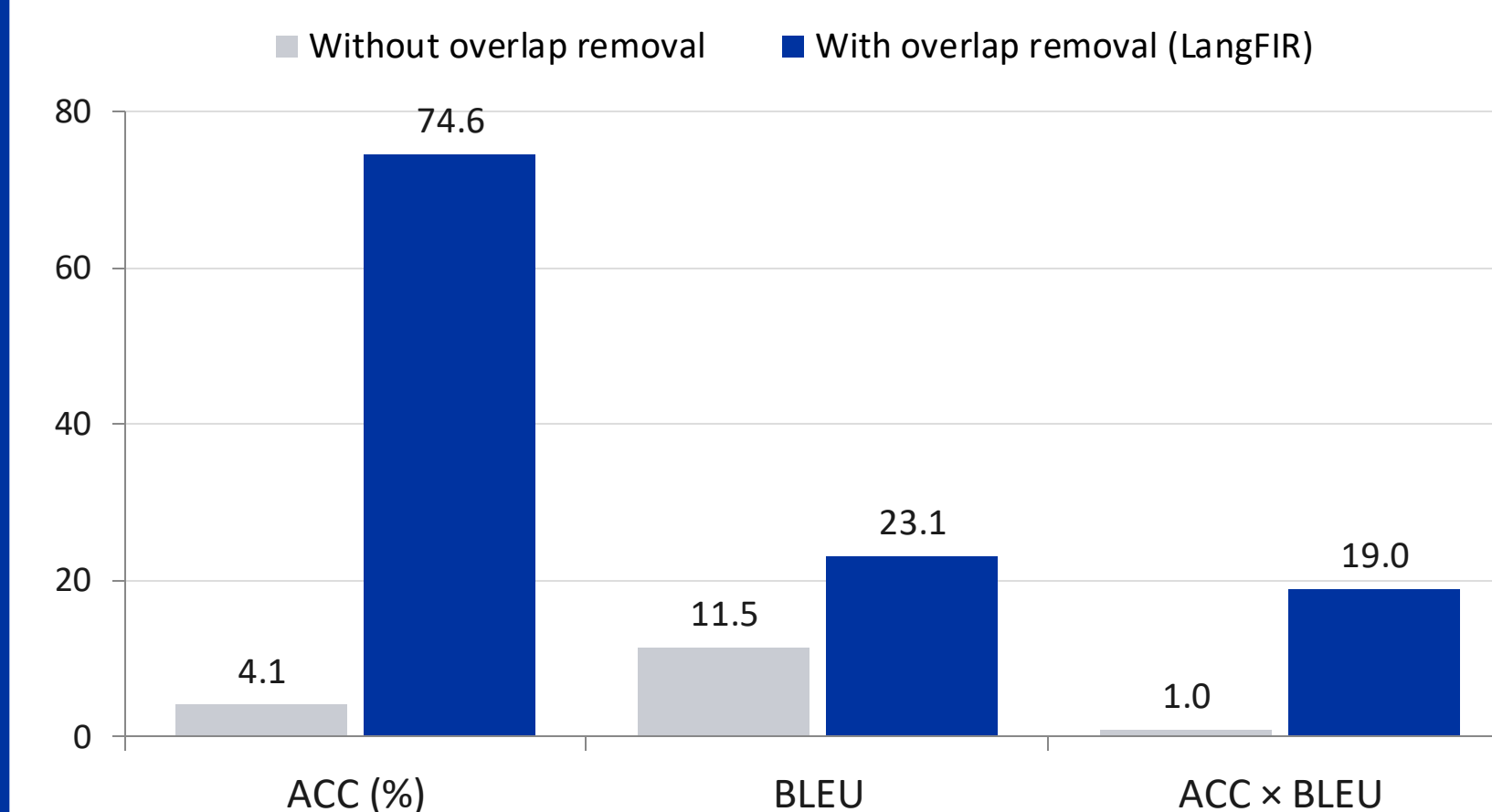
Sparse: < 5 features per language, stable by $N \approx 100$ **Selective:** top feature fires only on its own language



Causal: ablating the top-2 features raises cross-entropy (ΔCE) only for the target language (in color)

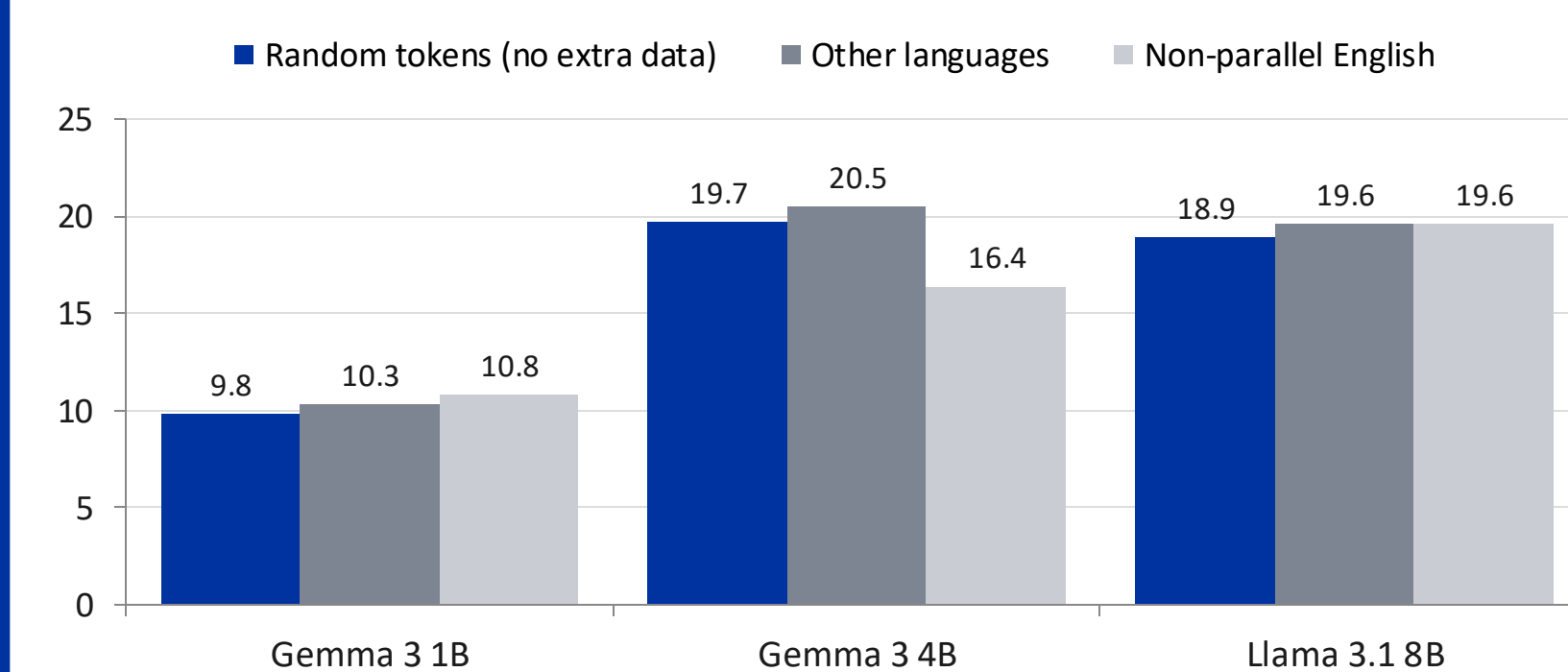
English is the exception: its features are weak and ablating them barely matters, likely because English is entangled with general concepts in English-centric LLMs.

The Filter Is Essential



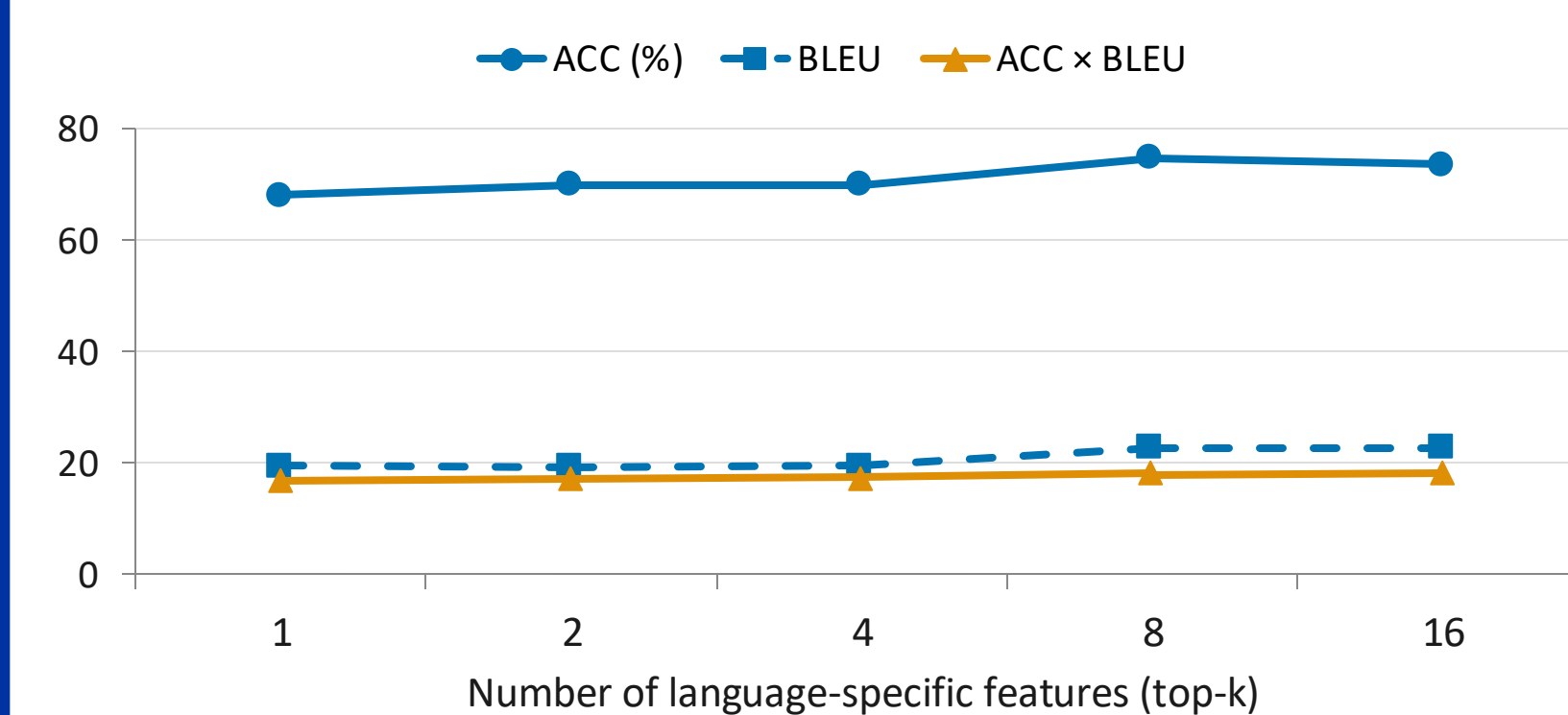
Skipping overlap removal (step 4) on Llama 3.1 8B degrades ACC by $\sim 18\times$, BLEU by $\sim 2\times$, and ACC $\times$ BLEU by $\sim 19\times$.

Random Tokens Suffice



Random tokens are competitive with filters that need extra data.

One Feature Is Enough



The top feature is enough to steer, and going from $k = 1$ to $k = 16$ improves only slightly.

Experimental Setup

Multilingual generation control task prompt

Translate an English sentence into a target language.
English: {source text} Target language:

Cross-lingual answer generation task prompt

Read the following passage and answer the question.
Context: ... Question: ... Answer:

Models: Gemma 3 1B, Gemma 3 4B, Llama 3.1 8B

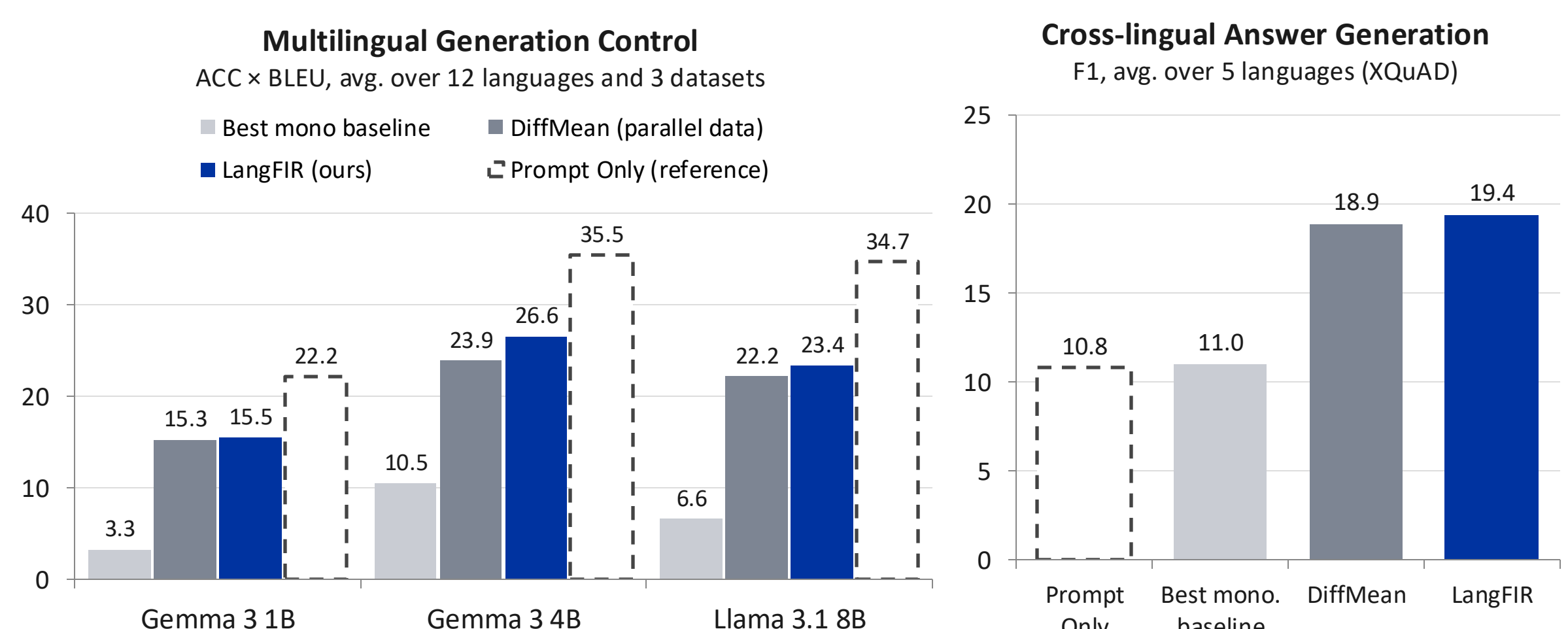
Languages: es, fr, de, it, id, pt, vi, pl, cs, tr, zh, nl

Data: FLORES+, WikiMatrix, Tatoeba, XQuAD

Metrics: ACC (translation accuracy), BLEU (translation quality), ACC $\times$ BLEU (composite score); F1

Baselines: 5 parallel-data and 2 monolingual steering methods

Steering Results



On multilingual generation control, **LangFIR is the best steering method on all 3 models using only monolingual data**. For cross-lingual answer generation (which has longer English contexts), prompting keeps outputs in English, whereas steering succeeds, with LangFIR performing best.

Using LangFIR in Practice

- $k = 1$ to 2 features already steer well.
- 10 to 100 monolingual sentences suffice.
- Use $\tau \geq 0.8$.
- Steer at middle layers.



Paper



Code



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